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Research Statement

My research is in stable homotopy theory, with connections to arithmetic and algebra. I study duality in algebraic K-theory and the algebraic structures arising in equivariant stable homotopy theory. These directions share an interest in how symmetry changes mathematical objects and their invariants. They also offer opportunities for undergraduate research beginning with familiar concepts in algebra and topology.

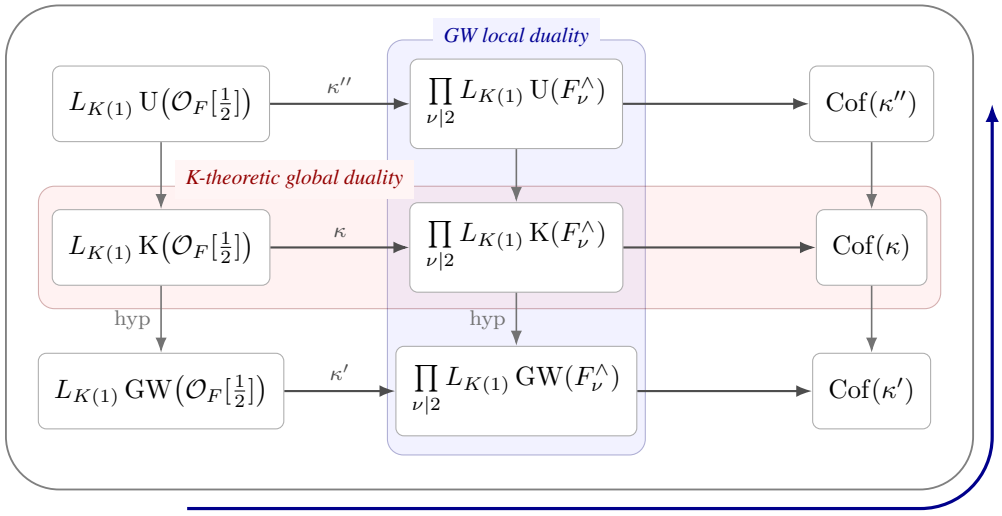
Duality in algebraic K-theory

Algebraic K-theory studies rings through invariants built from their projective modules. In number theory, classical dualities describe relationships among arithmetic cohomology groups. My research asks whether these relationships extend to algebraic K-theory itself. The goal is a duality of spectra, which captures both the individual groups and the homotopical structure connecting them.

My dissertation establishes K-theoretic Tate–Poitou duality at the prime 2, completing the global duality previously proved at odd primes by Blumberg and Mandell. The remaining case required accounting for the behavior of real embeddings. The theorem relates global K-theory to the term measuring the difference between global and local K-theory, while local K-theory is self-dual. The relevant notion of duality is Anderson duality, a homotopical analogue of taking the dual of an abelian group.

This result also applies to the K-theory of the sphere spectrum, a fundamental object in stable homotopy theory. The cyclotomic trace approximates this difficult invariant by one that is more accessible to computation. At the prime 2 and in nonnegative degrees, my theorem identifies the information missed by this approximation in terms of the arithmetic K-theory of the integers.

Real algebraic K-theory incorporates an order-two symmetry. Forgetting this symmetry recovers ordinary K-theory, while taking fixed points leads to Grothendieck–Witt theory (GW), which studies modules with nondegenerate bilinear forms. With Lucy Yang, I establish a compatible duality over 2-adic local fields for the sequence $\mathrm{U} \to \mathrm{K} \to \mathrm{GW}$. Our ongoing work seeks a global duality encompassing all three theories. In the diagram, the red row and blue column indicate the dualities already obtained. We expect the full diagram to be self-dual under a half-turn rotation.



My long-term goal is to understand duality and descent in Real algebraic K-theory of rings with involution. I conjecture that the arithmetic dualities above arise from a single Real duality. In ordinary K-theory, descent explains how arithmetic duality passes from cohomology to spectra. A corresponding equivariant

cohomology theory for Real algebraic K-theory is not yet available. I approach this problem from the other direction, using the expected duality to guide the search for a Real descent theory.

The sphere-spectrum application above connects arithmetic duality with the cyclotomic trace. This relationship motivates my study of Real trace methods as an approach to duality in Real algebraic K-theory. With Gerhardt, Keenan, Moreno, and Quigley, I develop homological methods for studying Real trace invariants together with their equivariant structure. We construct a multiplicative spectral sequence and carry out explicit calculations for Real bordism theories, extending the methods of Bruner and Rognes to the Real setting. A natural next step is a Real Segal conjecture for Real complex bordism and Real Brown–Peterson theory. This would advance our understanding of Real trace invariants and ultimately of Real algebraic K-theory of Real bordism theories.

Equivariant stable homotopy theory and algebra

Equivariant topology studies spaces together with group actions, such as rotations or reflections. Accounting for symmetry changes both the invariants of a space and the algebraic structures they carry. In equivariant stable homotopy theory, familiar operations such as addition and multiplication are accompanied by transfers and norms. Understanding which operations can coexist, and how they interact, is a central problem in equivariant algebra.

With Chan, Mehrle, Sanchez Ocal, Osorno, Szczesny, and Verdugo, I study this problem through transfer systems, combinatorial objects describing which transfers and norms are allowed. We ask whether compatible choices of these operations can be realized by operads governing their interaction. We prove realization when all additive transfers are available and establish further cases. We conjecture that every compatible pair can be realized, seeking a general correspondence between combinatorial compatibility and realizable algebraic structure.

With Noah Wisdom, I study how representation gradings organize the homotopy groups of equivariant spectra. For the order-two group, within the class of gradings modeled on stable homotopy groups, we show that the additional symmetries of representation degrees are already determined by the Mackey functor structure.

Undergraduate research I want students to learn to formulate their own mathematical questions. A starting point is a familiar definition: what changes if we weaken a condition, or combine two concepts usually studied separately? Equivariant rings offer an example, bringing together rings and symmetry to ask which definitions, examples, and properties are natural. The structures and invariants of manifolds with symmetry offer another direction. I would guide students in turning these broad questions into problems they can explore through concrete examples. As patterns emerge, we would refine their questions, formulate conjectures, and seek proofs or counterexamples. My aim is for students to experience how a question develops through investigation and to take increasing responsibility for the direction of their work.